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Towards Textile Templating in Mycofabrication: Knitted Structures as Substrate and Scaffold

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This study investigates the potential of knitted textiles as both substrate and three-dimensional scaffold for mycofabrication, contributing to the development of textile templating approaches in biofabrication. While textiles are commonly used as reinforcement in mycelium-based composites, their role as active, designable interfaces for growth remains underexplored. Two experimental approaches using Ganoderma lucidum were conducted: seeding-based inoculation and liquid-state fermentation. The first evaluated fibre substrates (namely acrylic, wool, cotton, and linen) on mycelial colonisation. Cellulosic fibres, particularly cotton and linen, supported more integrated and extensive growth, while wool enabled limited surface expansion and acrylic showed minimal colonisation. Overall, low biomass shows that textiles alone are insufficient nutrient sources, though fibre type influences moisture behaviour. The second experiment examined fabrication potential using submerged three-dimensional knitted structures. Cotton scaffolds showed more uniform colonisation and increased structural stability, whereas wool exhibited negligible growth. Growth patterns consistently followed areas of liquid exposure, emphasising the role of moisture and nutrient distribution and presenting potential transfer into textile design applications. The findings indicate that textiles function more effectively as moisture-regulating interfaces and structural guides rather than sole substrates for *Ganoderma lucidum*. By linking textile design parameters to mycelial behaviour, this research creates a basis for further development and highlights how computer numerical control (CNC) knitting technologies and textile design choices can expand opportunities in mycofabrication.

Keywords: Knitted Textiles; Mycofabrication; Textile Templating; Biofabrication

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1. Introduction

Textiles are widely employed in hybrid biofabrication due to their suitability as biological, physical, and mechanical interfaces. In tissue engineering, knitted scaffolds support biocompatibility and cell adhesion while enhancing mechanical strength and structural performance in bioassembly processes (Jiang et al., 2021; McCullen et al., 2010). Textiles also enable controlled moisture environments, acting as "fibre highways" that facilitate microbial translocation via water layers along fibres (Sherry et al., 2023). In widely used mycelium-based composite (MBC) solid-state fermentation (SSF) processes, lignocellulosic substrates such as wood chips provide essential nutrition and moisture for mycelial growth (Elsacker et al., 2020). While from a textile design perspective, opportunities are recognised for utilising textiles' tunability for biologically active interfaces for biodesign (Congdon, n.d.; Scott, 2018, 2023), mycofabrication research in particular has noticeably focused on textile as a reinforcement in common composite or mycelium leather applications (Rigobello et al., 2022; E. Williams et al., 2022; Womer et al., 2023). Only limited studies explore textiles beyond reinforcement, particularly as substrates or growth scaffolds in hybrid living materials, opening the space addressed by this study. Knitted textiles were selected because they enable precise control over material, structural, and environmental parameters through programmable fabrication (Scott, 2018; Jiang et al., 2021). Their tunable porosity, moisture behaviour, yarn allocation, and three-dimensional geometry make knitting particularly well suited to investigating how textile design can actively influence biological growth during mycofabrication. Computer numerical control (CNC) knitting further enables material and structural parameters to be digitally specified and translated into customised textile geometries. In this context, textile templating refers to the use of textile structures as active designed frameworks that guide biological growth through their material, structural, and moisture-regulating properties. This positions textiles not only as passive supports but as design variables influencing growth behaviour and material formation (Kaiser-Style and Scott, 2026).

2. Background

2.1. State of the Art

Current research in mycofabrication demonstrates that knitted textiles can support uniform mycelial colonisation when combined with supplementary nutritional hydrogel systems in laboratory settings (Helberg et al., 2019). Although design approaches increasingly integrate mycelial surface patterning with textile design techniques (Kaiser et al., 2024; N. Williams & Collet, 2021), the broader role of textile systems and their design potential for bioassembly remains underexplored.

Scaffold-led approaches, such as Mycelium-tectonics (Tabellini, 2015) introduced the colonisation of loose linen fibres within supportive laboratory setups. At a larger scale, the Liform project demonstrates a customised growth chamber enclosing a rope-based textile system to facilitate water supply for mycelial colonisation (Bryskina, 2017). These precedents highlight the untapped potential of tunable knitted structures as both substrate and three-dimensional scaffold for mycelium growth. Viewed through the lens of material intelligence, these precedents position textiles not simply as passive supports, but as designable interfaces that mediate moisture, structure and biological growth within living material systems. Understanding this interaction contributes to developing novel mycofabrication techniques based on textile templating.

2.2 Preliminary Experiments

A series of design-led exploratory tests were undertaken varying textile parameters, including loose fibres, knitted patterns, fabric construction and inoculation methods (Figure 1). A series of knitted fabrics were either submerged or dipped in a liquid containing mycelium and nutrient media. This process adapts the protocol by Elsacker et al. (2023). While the submerging technique produced growth beyond the textile scaffold, the dipping method resulted in more concentrated growth on the textile. Across exploratory tests, cellulosic fibres (flax, hemp, paper yarn, cotton) showed similar colonisation rates, while acrylic fibres exhibited limited growth. These observations (Figure 2) informed the isolation of key variables, particularly substrate types and fabrication method, examined in the following experiments.

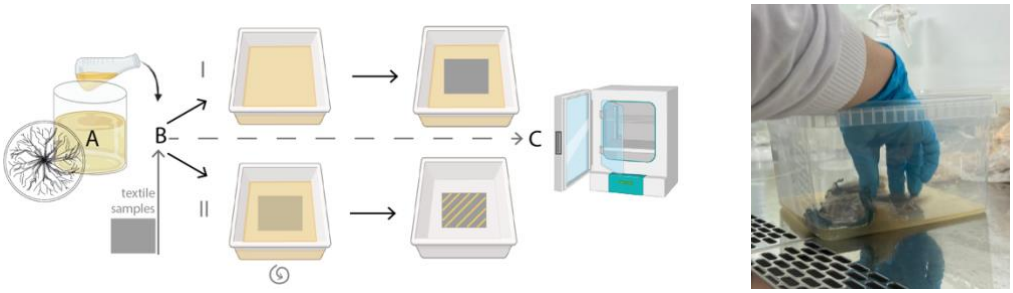


Figure 1. The process diagram (left) illustrates the inoculation workflow based on the liquid-state fermentation (LSF) technique (A), using two fabrication methods (B) for textile sample inoculation. BI shows the textile resting on nutrient media during growth, while BII depicts sample submersion (right) before transfer to an empty growth box. Both methods were followed by the same incubation procedure (C) for 15 days at 26°C in a static incubator.



Figure 2: Preliminary design-led tests under microscopy reveal varied mycelium–textile interactions and fabrication potential. A) Microscopic view of mycelial adhesion on loose flax fibre (close-up from B). C) Knitted cotton fabric immersed

in liquid culture using method I after 10 days (side view). High absorbency allows the loose cotton structure to retain substantial liquid medium, while oxygen availability is visible above the surface line, producing detailed growth that resolves individual loops. D) Filamentous mycelial network between loops (close-up from E). E) Mycelial hyphae exhibit climbing and binding behaviour between yarn formations in a loosely knitted 3D paper yarn structure fabricated with technique I. The sample consists of 100% paper yarn (Manila hemp, 2-ended, 2×1/21 Nm), manually formed into a looped 3D structure by suspending loops after 30 rows (×5). The material shows high liquid uptake, supporting filamentous hyphal growth, including bridging behaviour across gaps and the use of yarn structures as translocational pathways (F). G) Close-up of the fully colonised surface of sample IIa, demonstrating high-resolution replication of textile surface features. H) Detail of loop propagation in sample Ib (hemp yarn). I) Mycelial growth closely follows the loop structure of a cotton knit.

3 Aim and Methodology

The aim of this research was to investigate how knitted textiles could act as both substrate and three-dimensional scaffold for mycofabrication. This is explored in two ways:

- examining the relationship between knitted textile substrates and mycelial growth parameters,
- translating the relationship into a knit-specific fabrication approach.

The experiments combine controlled laboratory methods with an exploratory, design-led approach, enabling analysis of interconnected parameters such as growth performance, textile substrates, and resulting form and design features.

3.1 Leading Laboratory Protocol

This study adapts the protocol by Elsacker et al. (2023), which produces pure mycelium-based materials with leather-like properties through nutrient-rich liquid medium cultivation. The protocol exhibits reduced primordia formation, supporting the development of homogeneous mycelium skins suitable for leather-like materials (Elsacker et al., 2023). The liquid approach aligns with soft textile materials and was modified by incorporating knitted textile samples during the static incubation phase (Figure 1). This adaptation focuses on the influence of textile properties on key environmental growth parameters. All experiments use Ganoderma lucidum. A liquid mycelium culture was prepared using 2% malt extract broth (MEB).

Experiment 1 (Section 4) uses mycelium pellets (the initial stage before liquefaction) seeded directly onto textile substrates without additional nutrients.

Experiment 2 (Section 5) builds on the full liquid fermentation process, using a liquid mycelium–nutrient culture. The inoculated culture was prepared by incubating submerged mycelium pellets on a rotary shaker for five days, followed by static incubation to form mycelium skins. All experiments were conducted under sterile laboratory conditions using a fume hood, incubator, and filtered growth boxes. After 15 days (harvesting), the culture was reactivated by adding 2% fresh MEB and textile scaffolds for a further incubation cycle (Elsacker et al., 2023).

3.2 Textile Design

To evaluate suitable material combinations, four fibre types were selected, comprising both non-cellulosic and cellulosic yarns. The non-cellulosic group included acrylic (2/30 Nm, peach and forest green, supplied by Uppingham Yarns) and wool (2/30 Nm Merino worsted lambswool, midnight blue, supplied by Fairfield Yarns), with wool specifications aligned with those used in the Bioknit prototype tests (Agraviador et al., 2022). The cellulosic group consisted of cotton (2/30 Nm, 100% cotton, charcoal/grey, supplied by Uppingham Yarns) and linen (2/28 Nm, navy 935, supplied by Uppingham

Yarns). Samples were produced in 1 x 1 rib using a Shima Seiki 12gg SSR122 system or equivalent specifications on an 8gg Dubied knitting machine, depending on fibre compatibility.

4 Experiment 1: Viability potential of textile fibre substrate

4.1 Set-up

The viability of mycelial growth on textile substrates was evaluated using non-cellulosic yarns, acrylic (1) and wool (2) and cellulosic yarns, cotton (3) and linen (4) (as specified above).

Figure 3 illustrates the experimental setup, including preparation and seeding of mycelium pellets (Figure 3A) across two textile geometries (I and II; Figure 3C). Sample group I consisted of square samples (5 × 5 cm; 30 wales, 38 courses), while group II comprised rectangular lengths (2 × 16 cm; 20 wales, 105 courses). All samples were pre-soaked in 2 litres of deionised water for 72 hours and sterilised by autoclaving at 121°C for 30 minutes (Figure 3B). Due to size differences, group I samples were placed in sealed Petri dishes, while group II samples were incubated in closed growth boxes with air filters. For each material, four samples were incubated separately, while a fifth was included in a mixed-condition box containing one sample of each material, giving five replicates per material overall. Water uptake varied considerably due to differences in size, material, and fabric weight. Based on sample weight, group I provided approximately half the material mass of group II. Accordingly, average water uptake in group I was approximately half that of group II. Group I samples showed: acrylic 3.2 g, wool 3.6 g, cotton 3.4 g, and linen 4.1 g. Group II samples showed: acrylic 8.6 g, wool 8.7 g, cotton 7.4 g, and linen 9.3 g. Water uptake was proportional to dry weight increase but varied across fibre types.

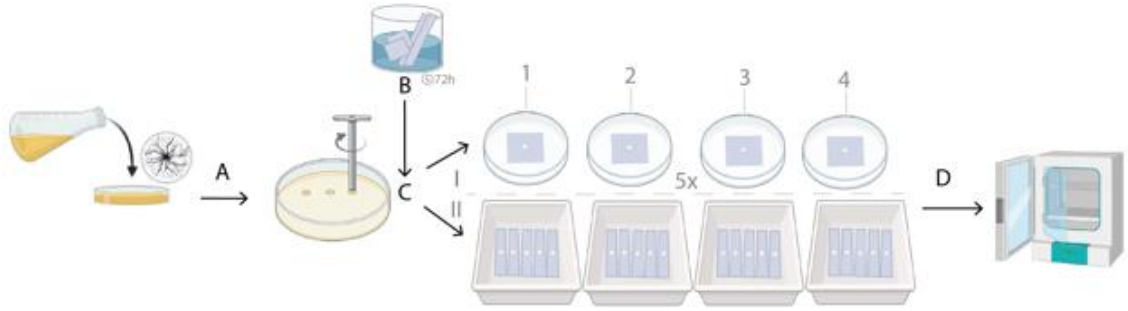


Figure 3. Schematic describing the experiment set-up. (A) Mycelium pellets (prepared according to Elsacker et al., 2023) were seeded in the middle of the 72 h water-pre-soaked (B) sterile textile samples. (C) Inoculated I samples were sealed in Petri dishes, while group II samples were positioned in closed growth boxes with a growth filter. (D) All samples were put in the incubator at 26°C and removed after 12 days. Icon creation supported through the use of Biorender.com

4.2 Growth process

Hyphal expansion was rapid during the initial days but slowed by day 6, likely due to limited humidity or accessible moisture. Growth was monitored until day 12 to assess further development. Figure 4 illustrates the growth process on wool samples. Day 2 (left) shows initial branching from the pellet centre (red cross), while day 6 (right) displays a circular network with denser hyphae at the advancing edges colonising the fabric surface. Growth success was assessed qualitatively based on observed development patterns.

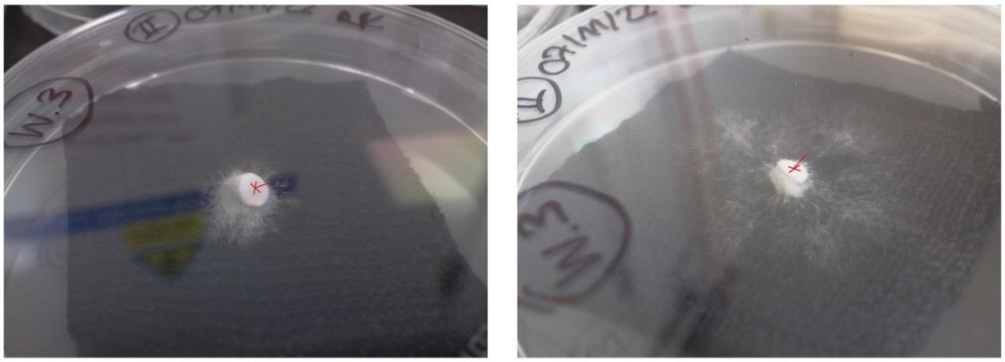


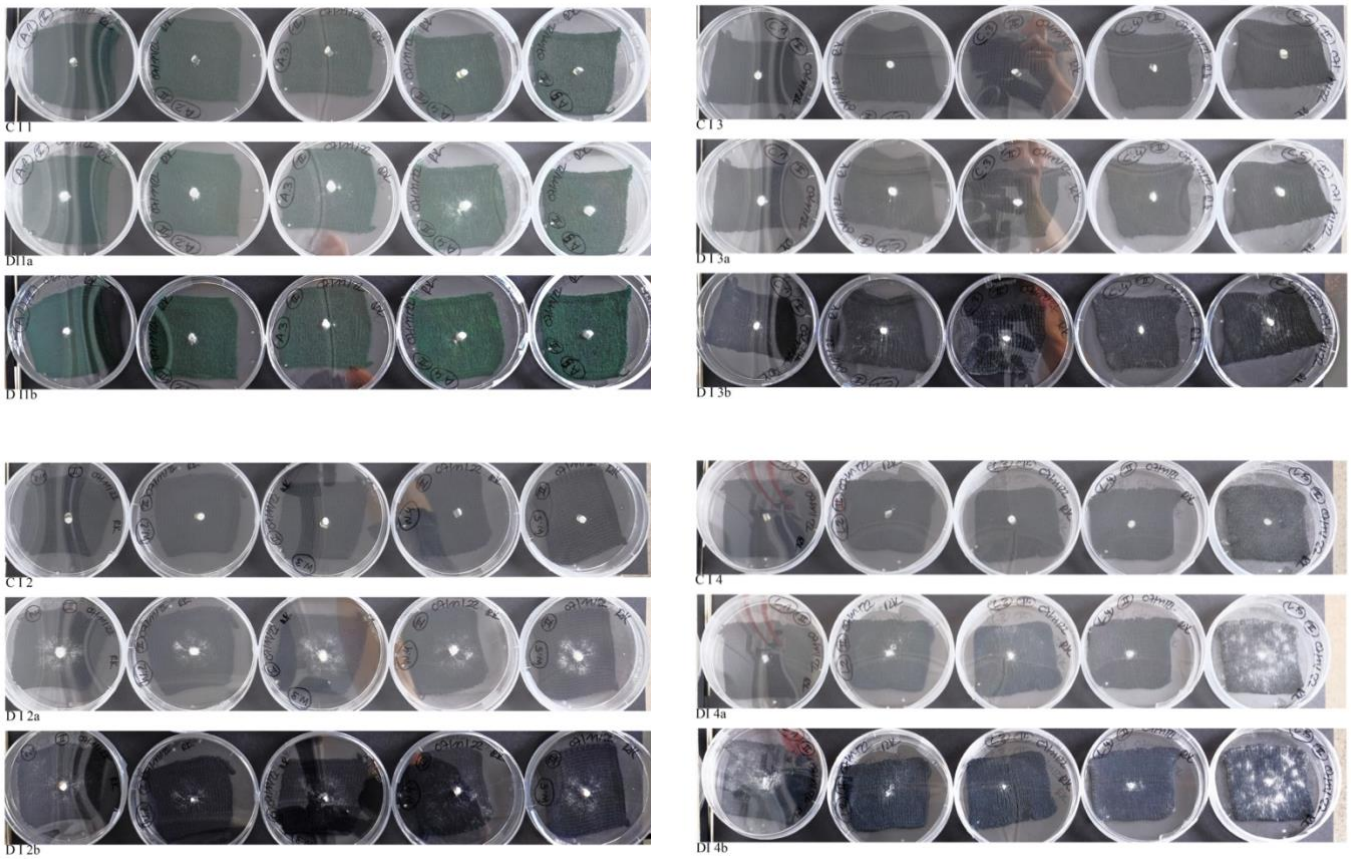
Figure 4. Fabric sample from category I.2.3 (third of five replicates; square wool sample, 5 × 5 cm). Close-up of hyphal development on day 2 (left) and day 6 (right), showing a circular branching pattern across the surface. The red cross marks the inoculation centre, while the red line indicates the orientation of the inoculation pellet, reflecting the slightly different perspectives in the images.

4.3 Results

Although overall growth was low, the experiments show that yarn type significantly influences colonisation patterns and morphology. Growth ranged from highest to lowest on linen, cotton, and wool, with no growth observed on acrylic beyond the seeding point (Figure 5). Acrylic samples (DI1, DII1 in Figures 5 and 6) exhibited minimal expansion from the inoculation centre.

Wool samples showed circular branching typical of mycelial colonisation, limited to the textile surface. In contrast, cotton and especially linen demonstrated more integrated growth within the textile structure (Figure 5: DI3, DI4; Figure 6: DII3, DII4). In cotton samples (DI3), the inoculation pellet lost its defined circular shape, with growth concentrated around the centre and extending toward the outer edges (Figure 5, DI3b.3). In DII3, fine linear branching developed along the length of the sample with clear spacing between strands. Linen samples (DI4) showed more integrated colonisation than wool, with some samples exhibiting randomly distributed speckled growth (DI4b.5) and the highest overall surface coverage.

In DII4, growth differed from the square samples, forming wide, dense branching across the stripe. A consistent pattern was observed, with a 3–4 cm gap after the inoculation centre followed by intense branching and thin-layer coverage toward the edges. Replicates showed consistent behaviour across all fabric categories (except as mentioned DI4b.5).



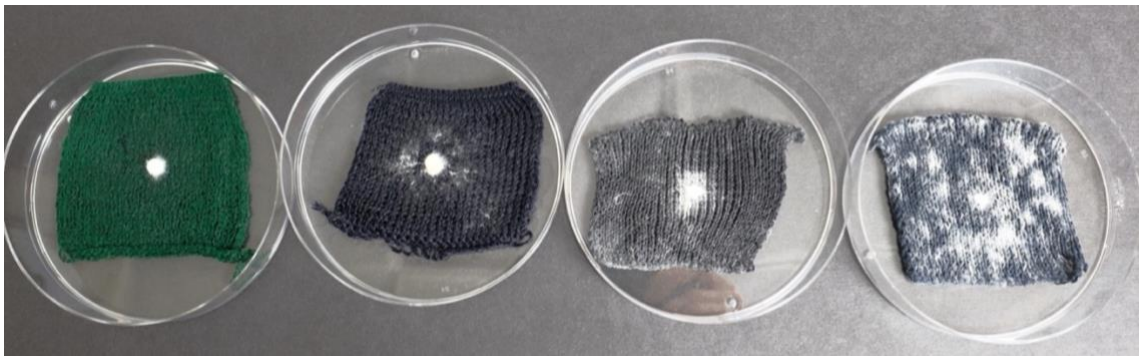


Figure 5. Top: Scans depict the result before (Cs) and after (Ds) growth in relation to yarn and size categories. Mycelial growth is visible in white on dark, coloured textile samples acrylic (1), wool (2), cotton (3) and linen (4). DI-x-a is presented after 6 days of growth; visuals exhibit loss of hyphae thickness intensity due to effects of drought stress, as visible in DI-x-b figures depicted on day 12, when growth was stopped. Below: Close-up after stopping the growth with an open Petri dish for better visibility, displaying the most intensely grown sample of category acrylic (1) sample DI1b.1, wool (2) DI2b.1, cotton (3) DI3b.3, and linen (4) DI4b.5 after 12 days of growth.

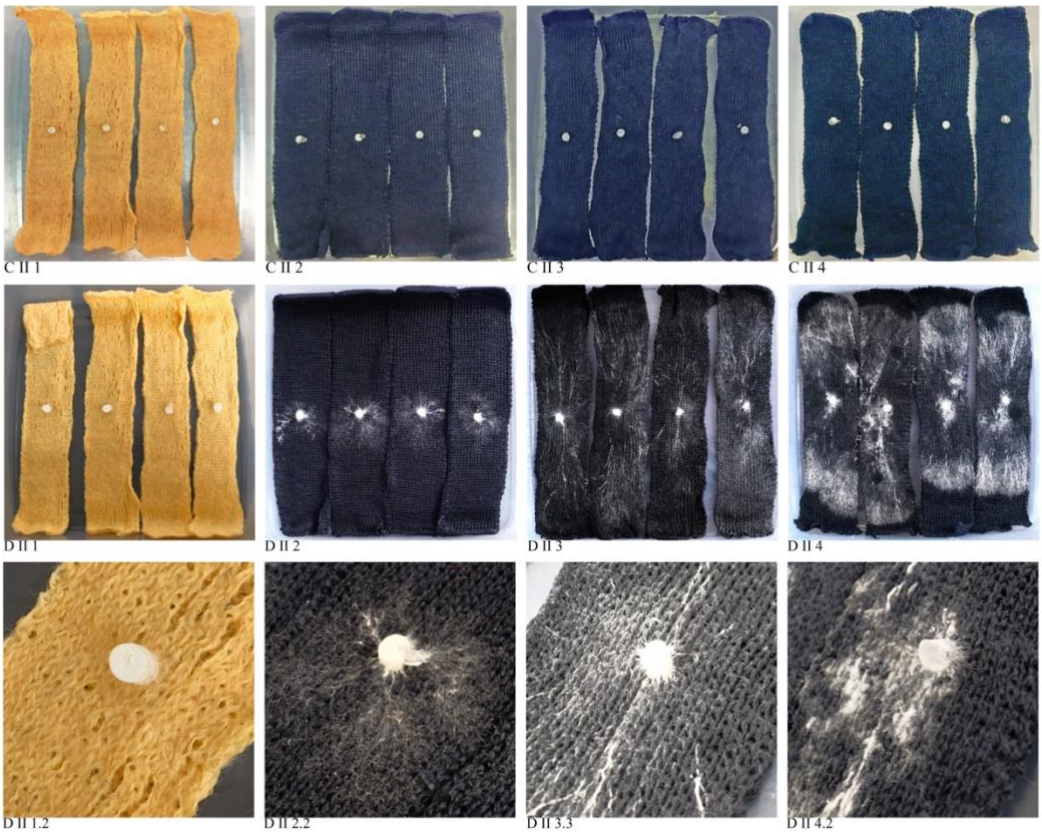


Figure 6. Overview of textile category shape II (stripes instead of squares) on growth day 0 after inoculation (Cs; top), and on day 12, the last day of growth (Ds; middle), displaying 4 out of 5 replicate textile stripes from left to right: acrylic (1), wool (2), cotton (3) and linen (4). The bottom section displays close-ups of one of the samples within the respective categories, highlighting the differences in growth development and integration around the centre inoculation point within the fabric fibres. Numbers of samples counting from left to right.

4.4 Discussion Experiment 1: Mycelium growth and fibre substrate

Limited colonisation indicates that textiles cannot serve as sole nutrient substrates and provide insufficient moisture storage. However, they function as moisture-active interfaces within the design process. Although samples remained damp after growth ceased, moisture content was not measured, and visual observations suggest drying, indicating potential dehydration and nutrient limitation. These findings show that soaking textile fibres alone does not create adequate growth conditions. Low fabric mass in samples I and II may have further constrained growth, which could improve with higher moisture and nutrient ratios.

In wool samples (DI2, DII2), hyphae formed circular networks radiating from the inoculation centre, characteristic of surface-based colonisation. Radial morphology maximises area coverage for nutrient absorption (Sabantina et al., 2019), suggesting nutrient limitation in this setup. This is particularly evident in the woollen stripes, where radial growth may indicate active nutrient searching. The advancing edges of these circular patterns show denser, finer hyphae compared to the more established strands observed in cotton and linen samples (DI3/DII3; DI4/DII4), suggesting possible

nutrient transport during continued foraging. This aligns with findings that mycelium grown on agar tends to exhibit radial expansion, while growth on wood is more longitudinal (Sabantina et al., 2019).

Rather than simply identifying suitable fibre types, these findings demonstrate how textile design parameters can be intentionally varied to influence biological behaviour. Textile design therefore becomes an active means of shaping mycelial development through material selection, structure, and moisture management.

Further microscopy is required to better understand hyphal integration within textile structures. Degradation testing could provide quantitative insight into substrate utilisation. It should also be noted that samples were not washed prior to sterilisation, and residual processing agents may have influenced growth conditions.

5. Experiment 2: 3D Scaffolding Fabrication Potential

5.1 Set-up

Textile Samples

Based on pre-tests, a fabrication setup using the submersion technique (Figure 1, II) was developed for two 3D knitted tubular columns using wool (2; dark blue) and cotton (3; grey) yarns as specified in *Section 3*.2. The wool sample was produced on a Shima Seiki 12gg SSR122, while the cotton equivalent was fabricated on a hand-operated 8gg Dubied machine.

Scaffold Support

A bespoke scaffold consisting of metal rods and two transparent plexiglass discs enabled suspension of the textile structure (Figure 7, left). The scaffold and textile columns were dimensioned to fit within a standard 2 L Fisherbrand tall-form beaker (FB33128; 240 mm high and 120 mm external diameter), providing a reference scale for the experimental setup. The suspended knitted columns were held under tension and varied in cross-section along their length. Textile samples were autoclaved at 121°C for 30 minutes without prior soaking and attached to the scaffold using matching yarn. As the scaffold could not be autoclaved, it was sterilised with ethanol and UV exposure in a fume hood for 20 minutes.

Protocol and Process

The liquid culture followed the protocol described in Section 3.1. Approximately 900 ml of inoculated culture was placed in a 2 L beaker, and each textile column was submerged for 5 minutes per side (Figure 7, second from left), absorbing approximately 25 g of liquid. After inoculation, samples were transferred to a plastic container to stabilise the scaffold within airtight growth bags (Figure 7, second from right). Samples were incubated at 26°C for 28 days to compensate for slower growth and drying conditions. Monitoring was limited by reduced visibility and condensation within the growth bags (Figure 7, right). Samples remained static throughout and were removed after the incubation period.



Figure 7. Biofabrication process including, left: sterile, suspended columns before submersion wool (2) and cotton right (3); second from left: submerging process for 5 minutes on each side in the liquid culture before turning to the other side; second from right: after inoculation, placed in a standing container and wrapped in growth bags to be put in the incubator; right: fogging after removal from the incubator for monitoring, creating challenges in visibility.

5.2 Results

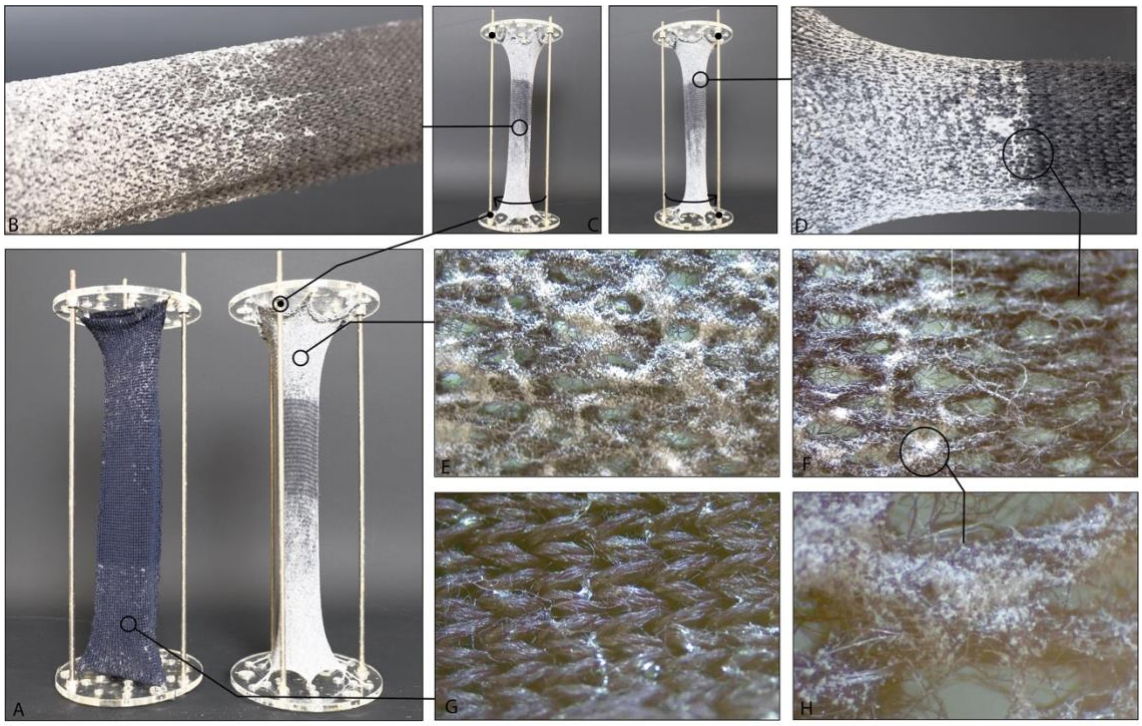


Figure 8. A) Mycelial growth on 3D knitted scaffolds after the growth period: wool (2, left) and cotton (3, right). B) Close-up of the lower fading line in the cotton column. C) Cotton column shown from different angles (±60° from A), highlighting variation in fading lines; the black dot marks the rod visible in A. D) Close-up of the upper fading line, showing a distinct cut rather than a gradient. E) Microscopic view of colonisation on the cotton fabric. F) Microscopic detail of D, illustrating the growth boundary. G) Microscopic view of colonisation on the wool fabric. H) Microscopic close-up of F, highlighting mycelial morphology and integration within the cotton fibre structure.

Figure 8 presents mycelial growth on the 3D knitted columns. Growth is visible on the cotton sample but does not form a continuous surface layer, while no growth is observed on the wool column (Figure 8A). The cotton column (3) shows even, integrated growth within the fibres in areas reached by the

liquid culture (Figure 8E). In contrast, the wool sample exhibits only residual mycelium and nutrient traces without active growth (Figure 8G).

Both columns display clear dipping marks corresponding to contact with the liquid culture. Due to limited liquid volume, submersion did not reach the central sections, resulting in uneven growth distribution. These areas remain uncolonised, forming visible gaps (Figure 8B–F). While growth was expected to bridge these zones, particularly in the cotton sample, no such development occurred.

The fading of the lower growth boundary suggests liquid flow along the column during inoculation/incubation, highlighting the role of water absorption and distribution in three-dimensional growth behaviour (Figure 8A, B).

After air-drying, the columns were removed from the scaffold. The wool sample collapsed due to the absence of growth, whereas the colonised cotton column exhibited increased stiffness and could remain upright, although the material remained structurally weak.

5.3 Discussion Experiment 2

Mycelium Growth and Fibre Substrate in a 3D Scaffold

Liquid nutrition enables more uniform growth distribution than local seeding, as shown in Experiment 1 (Section 4), and promotes greater integration within fibrous materials such as cotton (DII3 vs. Column 3), likely due to more even nutrient absorption and cotton's absorbency and moisture retention capabilities. However, the results highlight the need for further research to establish effective conditions for 3D mycelium skin cultivation. Optimising parameters such as nutrient supply, soaking duration, and continuous liquid delivery within the setup could improve colonisation outcomes. The hyphal morphology observed differs markedly from that of the previous experiment (Figure 6 vs. 8). Growth patterns suggest that aeration and oxygen availability in the 3D setup promote a crystal-like structure within the substrate (Figure 8H). This morphology, described as coral-shaped mycelium (Kuznetsova et al., 2020), is associated with aerial growth and aligns with observations by Elsacker et al. (2023) comparing floating and aerial hyphal formation in liquid cultures.

Fabrication Limitations and Scaffold Scaling

Conventional laboratory equipment proved inadequate even for the relatively small textile tubes used in Experiment 2. Scaling will likely require customised setups to maintain sterility and optimal growth conditions, particularly as liquid cultures are more susceptible to contamination than seeding or SSF methods (Rodriguez-Couto & Sanromán, 2006; Thomas et al., 2013). Additional challenges arose from the need for support structures to maintain textile form during submersion, complicating the fabrication of complex geometries under sterile conditions. These constraints suggest the need for dedicated incubator systems, such as customised chambers, as demonstrated in the Liform project (Bryskina, 2017) or in comparable approaches using textile scaffolds for bacterially driven biocalcification inoculated via spraying (Beyer et al., 2019). Overall, sterility requirements and fabrication constraints currently limit scalability and the practical application of this technique beyond controlled laboratory environments.

6. Discussion: Mycelium Growth Parameters and Textile Design Potential

The experiments demonstrate that textile design parameters, including fibre selection, knitted structure and moisture behaviour, can influence mycelial growth and therefore constitute active design variables within mycofabrication. The ability to tailor colonisation through these parameters is particularly relevant for knitting, given the potential for precise yarn allocation. The liquid dipping method integrates nutrients directly into the fabric without the bulk substrate typical of solid-state fermentation, enabling textile scaffolds developed through CNC knitting to support intricate geometries, including hollow forms.

This approach allows for the integration of diverse patterns, structures, and forms within scaffold design, including the incorporation of additional material systems and customised support setups. Bespoke yarn allocation further enables the application of biodesign strategies based on material choice, for example, ranging from reinforcement (Duhovic & Bhattacharyya, 2011) to temporary support systems, as used in tissue engineering (Jiang et al., 2021) within mycelial fabrication.

Liquid-based inoculation produced clearly defined growth boundaries, suggesting potential for precise pattern application. Figures 8A and 8D show distinct boundaries despite gravitational effects, while reduced intensity along the lower edge indicates a gradient associated with liquid depletion (Figure 8B). These gradients and fine lines reveal potentially designable effects, offering opportunities to create localised patterns and differentiated material properties within textile structures. Although not quantitatively measured here, the observed relationship between liquid exposure and colonisation suggests that porosity and moisture-related behaviour warrant further investigation as textile design variables influencing nutrient distribution and growth patterns. These effects may be transferable to established textile processes, such as screen printing on flat fabrics, where moisture-mediated nutrient delivery could function as an active design parameter. As moisture management is already well developed in sports and fashion textiles (Bal & Das, 2023; Venkatraman, 2015), these insights may inform future bioassembly systems through cross-disciplinary application. Figure 9 maps the identified relationships between growth behaviour and textile properties, highlighting their design potential for future textile templating developments.

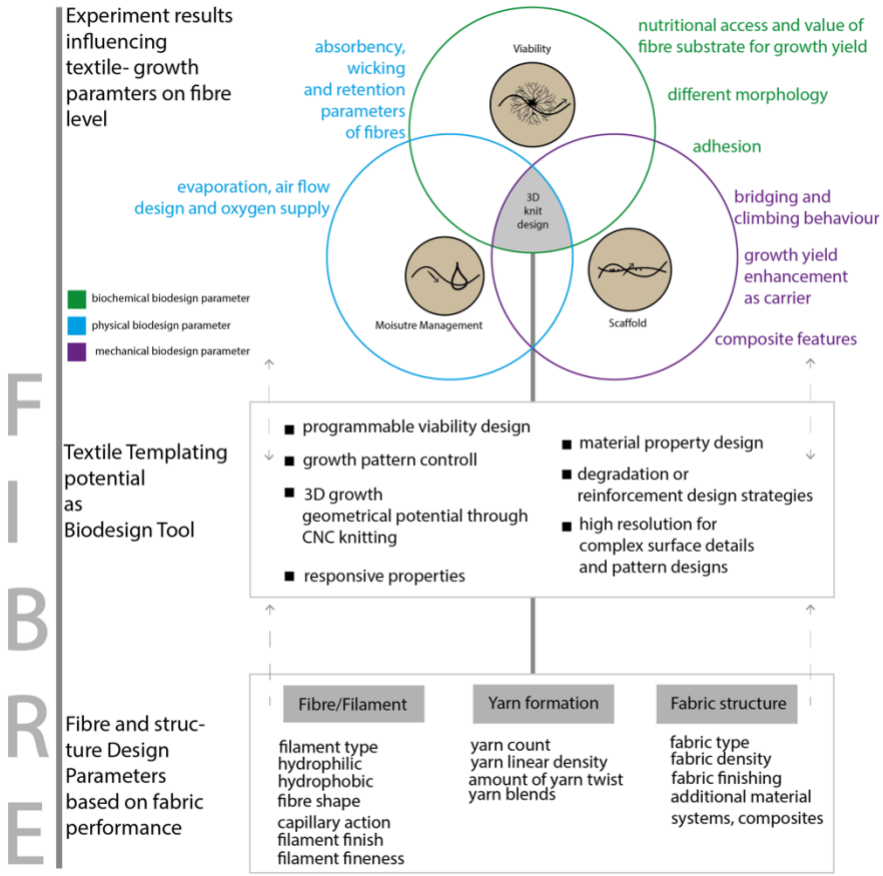


Figure 9. Mapping of interacting textile design parameters and mycelial growth behaviour. The upper section links experimental findings to textile-related growth parameters, while the lower section relates textile performance factors, particularly moisture management in sportswear (Venkatraman, 2015) to a broader understanding of design influences. Together, these relationships highlight the role of knitted textile design as a biodesign tool, illustrating its potential for textile templating (middle section).

7. Conclusion

Knitted textile formwork is becoming established as a key approach for scaling up mycelium composites, with researchers applying soft and flexible structures across a variety of large-scale applications. While textiles are commonly used as reinforcement in mycelium-based composites, their role as active, designable interfaces for growth remains underexplored. These experiments examined the relationship between textile substrates and mycelial growth to understand growth behaviour and fabrication potential. The findings show that biochemical, physical and mechanical properties associated with textile design parameters influence mycelial development and hyphal morphology.

In Experiment 1 (Section 4), mycelial morphology was shown to vary with fibre type. Cellulosic yarns such as cotton and linen supported more integrated growth, while wool enabled limited surface colonisation and acrylic did not support growth beyond the inoculation point. The findings suggest that textiles alone cannot sustain substantial growth under the tested conditions, but that fibre type affects moisture behaviour and colonisation patterns, highlighting its role as a design parameter within controlled environments.

Experiment 2 (Section 5) demonstrated how textile scaffolding parameters interact with liquid mycelial culture within three-dimensional structures. The method shows potential for generating intricate patterns and forms, particularly through CNC knitting. Further research is required to address challenges associated with scaling, material requirements, laboratory constraints and contamination risks as design experiments move beyond controlled environments.

Rather than treating textiles as passive reinforcement, this research demonstrates their potential to function as active design variables within mycofabrication. By experimentally linking textile parameters to mycelial behaviour, the study establishes an initial basis for textile templating as a textile design approach capable of shaping biological growth through material, structural and environmental design decisions. By revealing how textile properties and mycelial behaviour interact, the research contributes to an understanding of material intelligence within biofabrication and creates a platform for future developments. These include textile templates for mycofabrication using CNC knitting, as well as the transfer of this knowledge to other textile processes, such as liquid inoculation for print technologies. It is significant that mycelium biotechnology is considered from a textile perspective and that laboratory-based experimentation is underpinned by textile knowledge and expertise. The ability of textiles to operate at the intersection of design, biotechnology and mycology is increasingly important as the field transitions from fundamental science towards real-world applications.

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